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Vortex-induced vibration analysis of long-span bridges with twin-box decks under non-uniformly distributed turbulent winds

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ABSTRACT

With the increase of span length, long-span bridges become more flexible and susceptible to vortex-induced vibrations (VIV). Large-amplitude VIV may cause fatigue damage to structural components, and therefore accurate predictions of vortex-induced responses (VIR) are important. This paper proposes a mode-by-mode VIV analysis method for long-span bridges with twin-box decks under non-uniformly distributed turbulent winds. A semi-empirical model of vortex-induced forces (VIF), developed by the authors and validated by direct force measurements on an elastically-mounted twin-box section model under turbulent winds, is embedded in the VIV analysis method. The proposed analysis method is then used to analyze the VIV of a real long-span suspension bridge of a twin-box deck. The computed VIR of the prototype-bridge is finally compared with the VIR measured on-site by a structural heath monitoring system installed in the bridge. The comparative results show that the proposed VIV analysis method can be used to predict the VIR of long-span bridges with twin-box decks. Further studies also show that the non-uniform distribution of mean wind speed reduces the VIV of the bridge. Turbulence also diminishes the VIV of the bridge, and the reduction effect is larger on lower-order modes of vibration than on higher-order modes of vibration.

1. Introduction

Vortex-induced vibration (VIV) is a typical vibration caused by the interaction of bridge motion with incoming wind. Although VIV does not directly cause a bridge to collapse, it may induce fatigue damage to crucial structural components of the bridge and impair the driving safety of vehicles. Many long-span cable-supported bridges once suffered VIV, which include the Second Severn Crossing in the UK (Macdonald et al., 2002), the Deer Isle Bridge in the US (Kumarasena et al., 1991), the Great Belt Bridge in Denmark (Larsen et al., 2000), and the Xihoumen Bridge in China (Li et al., 2011).

With the further increase of span length, long-span bridges will become more flexible and susceptible to VIV. VIV with small amplitudes has to be allowed for some long-span bridges in practice as long as the safety and serviceability of the bridges are not seriously affected. This is particularly true for long-span bridges with twin-box decks. Therefore, accurate predictions of vortex-induced responses (VIR) of the bridges become important and necessary.

The currently available prediction methods of VIV are mainly based on wind tunnel tests of aeroelastic models of bridge deck sections. These methods generally fall into two categories. The first category directly uses the results from sectional model tests to predict the VIR of the prototype bridge. Ehsan and Scanlan (1990) proposed a semi-empirical model of vortex-induced force (VIF) of single-degree-of freedom (SDOF), and by assuming that VIV is harmonic in the state of lock-in, the maximum amplitude of VIR was derived from the semi-empirical VIF model as the function of several parameters. Once these parameters are identified from wind-tunnel test results, the maximum amplitude of VIR can be determined. Irwin (1998) also deduced the relationship between the maximum displacement of the wind tunnel test model and that of the prototype bridge. These methods are simple but VIF acting on the bridge, which is important for fatigue analysis, cannot be directly obtained by these methods. The second category uses wind tunnel test results to identify the parameters involved in semi-empirical models of VIF to obtain VIF time-histories, and then applies the identified VIFs to the analytical or numerical model of the bridge for VIR prediction (Ehsan and

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Scanlan, 1990). Barhoush et al. (1995) used a 2DOF semi-empirical VIF model, a finite element (FE) beam model, and the Newmark-Beta method to obtain the solution for VIR. Diana et al. (2006) solved the VIV problem of a bridge deck using a FE bridge model and a 2DOF semi-empirical VIF model. Considering the fact that the VIV of a bridge is always excited by one mode at a time, Meng et al. (2015) proposed a mode-by-mode solution method for VIV of the prototype bridge based on a nonlinear SDOF VIF model (Zhu et al., 2013).

Although it is a common practice to predict the VIV of the prototype bridge using wind tunnel tests on the sectional model as discussed above, the accuracy of these methods is not clear and the references concerning the comparison between the analytical and field measurement results are hardly found. It is speculated that three issues may hinder accurate predictions of the VIR of the prototype bridge. First, VIV is sensitive to structural damping but it is very difficult to accurately measure the damping of a bridge, especially the nonlinear damping of a bridge of large-amplitude vibration. Second, a certain level of turbulence always exists on bridge site and the studies on the turbulence effects on VIV are few. Last, for long-span bridges, incoming wind is not uniform. The mean wind speed as well as turbulence intensity may vary along the bridge axis (Li et al., 2011), but the existing VIV analysis methods cannot take into account the nonuniformity of the incoming wind. These issues need to be solved in order to achieve an accurate prediction of VIV for long-span bridges.

This paper thus aims to develop a mode-by-mode VIV analysis method for long-span bridges with twin-box decks under non-uniformly distributed mean and turbulent winds along the bridge deck. A semi-empirical model of vortex-induced forces (VIF), developed by the authors and validated by direct force measurements on an elastically-mounted twin-box section model under turbulent winds, is embedded in the VIV analysis method. The proposed analysis method is used to analyze the VIV of a real long-span suspension bridge of a twin-box deck. The computed VIR of the entire-bridge is then compared with the VIR measured on-site by a structural heath monitoring system installed in the bridge to validate the proposed method. The validated model is finally used for parametric studies.

2. VIF model for twin-box decks under turbulent wind

The governing equation of a sectional model of a bridge deck used in wind tunnel test under vertical VIF can be written as

$$m\left(\ddot{y}+2\zeta\omega_1\dot{y}+\omega_1^2 y\right)=F_{VI} \tag{1}$$

where m is the mass per unit length; ω_1 is the structural circular frequency; $\zeta = c/2m\omega_1$ is the mechanical damping ratio; c is the structural damping; y is the vertical displacement of the deck sectional model; the over-dot denotes differentiation with respect to time; and F_{VI} is the VIF.

Eq. (1) can be rewritten into a dimensionless form as

$$\ddot{\eta}(s)+2\zeta K_1\dot{\eta}(s)+K_1^2\eta(s)=\tilde{F}_{VI} \tag{2}$$

where $\eta = y/D$ is the dimensionless vertical displacement; D is the deck depth; $K_1 = \omega_1 D/U$ is the reduced natural frequency; U is the mean wind speed perpendicular to the bridge deck; $s = Ut/D$ is the dimensionless time; the over-dot here denotes differentiation with respect to the dimensionless time; and $\tilde{F}_{VI}$ is the dimensionless VIF.

The VIF model, developed by the authors and taken into account the turbulence effects (Zhu et al., 2017b), is used in this study and it is expressed as

$$\tilde{F}_{VI}=\frac{\rho D^2}{m}\left[Y_1(K)\cdot\left(1-\varepsilon_1(K)\eta^2-\varepsilon(K)\dot{\eta}^2-\varepsilon_t w_r^2\right)\cdot\dot{\eta}+Y_2(K)\eta+\frac{1}{2}\tilde{C}_L\sin(Ks+\varphi)\right] \tag{3}$$

where $Y_1(K)$, $Y_2(K)$, $\varepsilon(K)$, $\varepsilon_1(K)$, $\tilde{C}_L$ and φ are the parameters to be determined by wind tunnel tests under smooth incoming wind flow; ε_t is the to-be-determined parameter related to the turbulence-induced nonlinear damping term; $w_r(t) = w(t)/U$ is the dimensionless vertical turbulence component; $w(t)$ is the vertical turbulence speed; and ρ is the air density.

The proposed VIF model includes all the non-conservative motion-induced force terms. The validity of the proposed VIF model was examined using a newly-developed wind tunnel test technique that can obtain the VIF time-histories on an elastically-mounted sectional model. The parameters in the VIF model were identified with the VIF time-histories extracted from the measured total force time-histories. The term $m_r Y_1\varepsilon_1\eta^2\dot{\eta}$ was found to be very small and has a marginal influence on the VIR, where $m_r = \rho D^2/\text{m}$. The pure-vortex-excitation term $\tilde{C}_L\sin(Ks+\varphi)$ is also of negligible importance to the steady-state VIR, as discussed in Ehsan and Scanlan (1990) and verified by Zhu et al. (2017b). As a result, this VIF model can be further simplified as

$$\tilde{F}_{VI}=\frac{\rho D^2}{m}\left[Y_1(K)\cdot\left(1-\varepsilon(K)\dot{\eta}^2-\varepsilon_t w_r^2\right)\cdot\dot{\eta}+Y_2(K)\eta\right] \tag{4}$$

3. Mode-by-mode VIV analysis of the bridge under non-uniformly distributed winds

With the proposed VIF model, the governing equation of the mode-by-mode VIV of the entire bridge can be written as

$$\ddot{\eta}(s)+2\zeta K_1\dot{\eta}(s)+K_1^2\eta(s)=\frac{\rho D^2}{m}\left[Y_1(K)\cdot\left(1-\varepsilon(K)\dot{\eta}^2-\varepsilon_t w_r^2\right)\cdot\dot{\eta}+Y_2(K)\eta\right] \tag{5}$$

This is based on the assumption that the deck-dominated lower-order natural frequencies are well separated and that lock-in vibrations only occur at a single deck-dominated mode each time. Furthermore, the lock-in vibrations can be considered fully correlated (Zhu et al., 2017a). In such cases, the non-dimensional vertical displacement η at the span-wise location x of the bridge deck can be expressed as

$$\eta(x,s)=\Phi(x)v(s) \tag{6}$$

where $\Phi(x)$ is the non-dimensional mode shape of the bridge deck for the VIV; and $v(s)$ is the generalized coordinate.

By multiplying $\Phi(x)$ on both sides of Eq. (5) and conducting an integration along the bridge axis, the generalized governing equation, considering the non-uniform distribution of mean wind speed as well as turbulence along the bridge axis, can be written as

$$\begin{aligned}&M\left[\ddot{v}(s)+2\zeta K_1\dot{v}(s)+K_1^2 v(s)\right]\\&=\rho D^2\Big[\dot{v}(s)\textstyle\int_0^L Y_1(K)\Phi^2(x)dx-\dot{v}^3(s)\int_0^L Y_1(K)\varepsilon(K)\Phi^4(x)dx\\&\quad-\varepsilon_t\dot{v}(s)\textstyle\int_0^L Y_1(K)w_r^2(x,s)\Phi^2(x)dx+v(s)\int_0^L Y_2(K)\Phi^2(x)dx\Big]\end{aligned} \tag{7}$$

$$M=\int_0^L m\Phi^2(x)\mathrm{d}x \tag{8}$$

where M is the generalized mass; and L is the total length of the bridge deck. It is noted that in the cases that mean wind speed and turbulence vary span-wisely, not only the turbulence $w_r(s)$, but also the parameters Y_1, Y_2 and ε are the functions of x as they are functions of the mean wind speed.

Without considering the non-uniformity of the incoming wind, Eq. (7) can be reduced to

$$\begin{aligned}&M\left[\ddot{v}(s)+2\zeta K_1\dot{v}(s)+K_1^2 v(s)\right]\\&=\rho D^2\Big[Y_1(K)\dot{v}(s)\textstyle\int_0^L\Phi^2(x)\mathrm{d}x-\varepsilon(K)Y_1(K)\dot{v}^3(s)\int_0^L\Phi^4(x)\mathrm{d}x\\&\quad-\varepsilon_t Y_1(K)w_r^2(s)\dot{v}(s)\textstyle\int_0^L\Phi^2(x)\mathrm{d}x+Y_2(K)v(s)\int_0^L\Phi^2(x)\mathrm{d}x\Big]\end{aligned} \tag{9}$$

Both Eqs. (7) and (9) can be solved mode-by-mode with the Newmark-beta method for the VIR of the bridge.

4. Xihoumen Bridge and VIV events

4.1. Xihouman Bridge and structural health monitoring system

The Xihoumen Bridge is a suspension bridge located on the east coast of China with a main span of 1650 m and two side spans of 578 m each (see Fig. 1). The north side span is supported by suspenders but the south side span of the bridge is supported by a series of piers. This long-span bridge suffers strong typhoons and monsoons. As a result, a twin-box deck is adopted for its better performance against flutter. However, the geometry of the deck makes the bridge susceptible to VIV at low wind speeds.

The twin-box steel deck of the bridge is 36 m wide and 3.51 m in depth. The gap between the two box girders is 6 m. The north and south towers of the bridge are made of reinforced concrete and with a height of 236.5 m. The main cables of the bridge are composed of parallel steel wires.

The bridge crosses a narrow water way that lies between two islands and the bridge sits near the north island. The different terrains at the north and south ends of the bridge result in large variation of the wind characteristics along the bridge axis (Li et al., 2014).

A comprehensive structural health monitoring system has been installed on the bridge to monitor its performance and safety. The system was implemented in 2009 and then extended and upgraded in 2010. Monitoring wind and wind effects on the bridge is a major objective of the system. A detailed introduction to the system can be found in the work of Li et al. (2011, 2014). Some important information on the sensors regarding the work in this paper is presented here. Six three-dimensional ultrasonic anemometers were installed on the poles located at the middle and quarter points of the central span (see Fig. 1a). The anemometers were installed 6 m above the deck surface. As the middle anemometers were very close to and therefore strongly interfered by the main cable, only the winds recorded by the northern and southern quarter-span anemometers are used for this study. Three sets of accelerometers were also installed at the middle and quarter points of the central span (see Fig. 1b). Each set contains one accelerometer that records the lateral acceleration of the deck and another two on both sides of the deck to measure the vertical acceleration.

4.2. FE model of the bridge and its dynamic characteristics

The finite element (FE) model of the Xihoumen Bridge was established using ANSYS. The twin-box deck of the bridge was modelled as two parallel longitudinal beams connected by cross-beams. Both the longitudinal and cross beams were modelled with 3-D elastic beam elements with tension, compression, torsion, and bending capacities. Additional mass resulting from the secondary components of the deck was added to the model with mass elements. The towers were also modelled with 3-D elastic beam elements. The main cables and the suspenders were modelled with 3-D tension-only truss elements. The main cable and the towers were connected at the tower tops with coupled translational degrees of freedom. The bottom of the towers and the anchorages of the main cables were modelled as fixed ends. (see Fig. 2).

The established FE model can effectively capture the dynamic property of the prototype bridge. A comparison between the analytical and field-measured modal frequencies of the first 6 vertical modes, which are used for the VIV analyses of this study, is conducted and the results are listed in Table 1. The differences between the analytical and field-measured modal frequencies are quite small. For the 3rd and 6th vertical modes, which are the main focus of this study, the differences are below 0.6%.

4.3. VIV events recorded

Thirty-seven observed VIV events were reported in the work of Li et al. (2014). Seven of these events occurred with the 3rd vertical bending mode of the bridge with a frequency of 0.183 Hz and a structural damping ratio of 0.5% identified. Twenty-three of these events occurred with the 6th vertical bending mode of the bridge with a frequency of

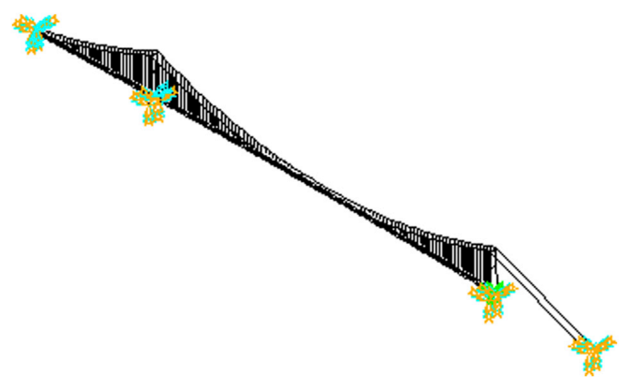


Fig. 2. FE model of the Xihoumen Bridge.

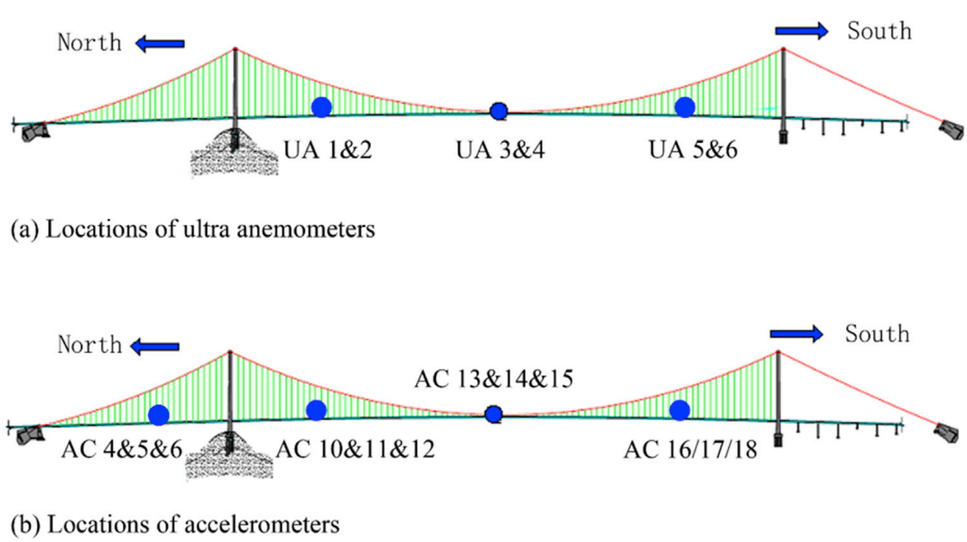


Fig. 1. The Xihoumen Bridge and the locations of the anemometers and accelerometers.

Table 1
Measured and analytical vertical modal frequencies.

Vertical mode order	Measured Freq.(Hz)	Analytical Freq.(Hz)	Diff.[a](%)
1	0.095	0.097	2.11%
2	0.133	0.133	0.00%
3	0.183	0.182	−0.55%
4	0.230	0.228	−0.87%
5	0.276	0.260	−5.80%
6	0.324	0.323	−0.31%

[a] Diff refers to the relative difference between the analytical and measured frequency.

0.324 Hz and a structural damping ratio of 0.42% identified. Although VIVs also took place in other modes, the data may be insufficient to give the whole picture of the VIV for those modes. As a result, only the on-site data regarding the 3rd and 6th vertical modes are used in this paper for comparison with the analytical results. The first six vertical mode shapes of the bridge are plotted in Fig. 3.

The events regarding these two modes are summarized in Table 2. The events E1-E7 are the VIV events associated with the 3rd vertical mode, while the events E8-E30 are the VIV events related to the 6th vertical mode. The event E1 captures the largest VIV recorded on the 3rd vertical mode and it is also the event observed with the largest displacement response among all the events. The event 17 is the largest VIV recorded on the 6th vertical mode and it is also the one with the largest acceleration response among all the events.

The time-histories of the incoming wind were simultaneously recorded with those of the responses during all the VIV events. The time-history and the power spectrum the incoming vertical turbulence of a wind data sample is plotted in Fig. 4. It can be seen that the dominant frequency range of turbulent wind energy is very low.

It can be seen from Table 2 that the mean wind speeds recorded by the north and south anemometers are different. Although the differences are less than 1 m/s in most cases, such non-uniform distribution of the mean wind speed may affect the VIR because VIV is very sensitive to mean wind speed. The differences between the turbulence intensities (I_u) recorded by the north and south anemometers are quite large. The difference in the recorded turbulence intensities can largely attribute to the topography at the bridge site as there is a hill on the south of the bridge (Li et al., 2014). The non-uniform distribution of the turbulence may also affect the VIV of the bridge.

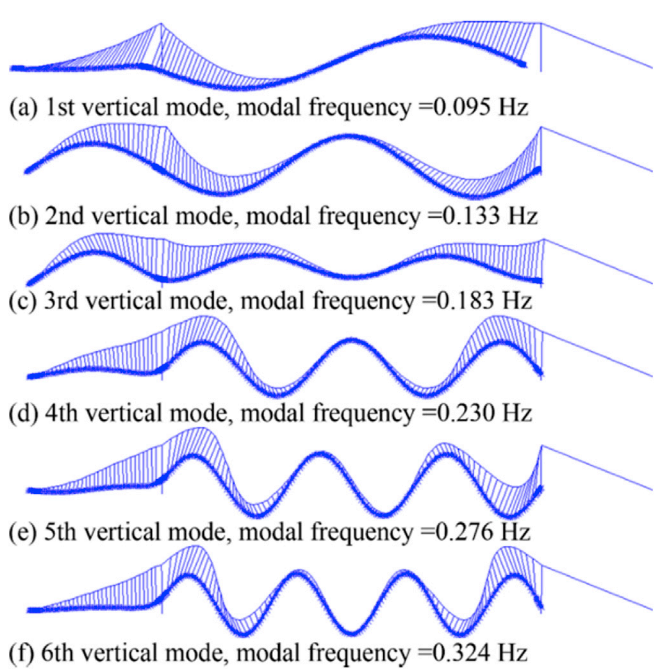


Fig. 3. First 6 vertical bending mode shapes of the Xihoumen Bridge.

5. Wind tunnel tests on Xihoumen bridge deck

Wind tunnel tests on the sectional model of the Xihoumen bridge deck were carried out in the TJ-3 boundary layer wind tunnel of the State Key Laboratory for Disaster Reduction in Civil Engineering at Tongji University, China (see Fig. 5). The twin-box deck of the Xihoumen Bridge was simulated using a sectional model with a scale ratio of 1:20. The model is 3.6 m in length, 1.8 m in width (with a 0.3 m slot between the two boxes), and 0.175 m in depth.

A newly-developed wind tunnel test technique was used to obtain the VIF time-histories on the spring-mounted twin-box section model (Zhu et al., 2017a, 2017b). The plan layout and details of the spring-mounted sectional model is shown in Fig. 6. An inner rigid frame with two longitudinal beams and four cross beams is used to provide sufficient stiffness. 7 pairs of force-measuring strips are embedded in the model. Each strip is made of light polymeric foam and some auxiliary parts that allow the strip to be connected to the inner rigid frame through a force balance. Apart from the 7 pairs of force-measuring strips, wooden plate skins are used to cover the rest of the model surface to form the completed twin-box section model. Totally 14 force balances are installed in the model. They directly connect the force measurement strips to the inner rigid frame. Marginal gaps of 1 mm width between the polymeric foam strip and the wooden-plate skinned rest of the model are retained so that the forces recorded by the force balances are only the inertia and aerodynamic forces acting on the light foam strip, each of which weighs only about 0.094 kg. The use of narrow force-measuring strips can largely reduce the proportion of inertia forces in the measured data and therefore improve the accuracy of acquired VIFs.

The lock-in range and the maximum displacement responses at different reduced wind speeds of the section model are depicted in Fig. 7. The lock-in range of the model under smooth wind flow is within the reduced wind speeds U_r from 6.03 to 8.82, where $U_r = U/fD$, f is the frequency, and D is the depth of the deck. The maximum dimensionless oscillation amplitude appears at U_r of 7.08, with the value of 0.059.

By subtracting the inertial and non-wind-induced force from the recorded force time-histories, the time-histories of the VIF can be acquired. The parameters described in Eq. (4) can then be identified from these VIF time-histories. The identified parameters of the VIF acting on the Xihoumen bridge deck are listed in Table 3.

6. Mode-by-mode VIV analysis results and comparison with field measurement data

6.1. Analyses with uniformly distributed smooth incoming wind

With the identified VIF and the established FE model of the bridge, the mode-by-mode VIV analysis of the Xihouman Bridge can be performed for various cases. This subsection concerns the VIV of the bridge under uniformly-distributed smooth incoming wind without turbulence. The analytical results of the vortex-induced maximum displacement response due to uniformly distributed smooth incoming wind ($I_w = 0$) are displayed in Fig. 8 for the first few vertical bending modes. Each lock-in range shown in Fig. 8 belongs to one vertical deck bending mode of the bridge. It can be seen that the vortex-induced maximum displacement responses of the bridge on mode 2 and mode 5 are relatively small within a relatively small wind speed range. With the increase of the number of mode, vortex-induced lock-in range becomes larger and larger. Theoretically, VIV can occur at all the vertical deck bending modes so that all possible modes that are susceptible to VIV within reasonable mean wind-speed and turbulence intensity ranges should be investigated to ensure the performance of a bridge against VIV. However, for the bridge investigated in this study, only the VIV events on the 3rd and 6th vertical modes were observed on site so far. As a result, only the vortex-induced responses on the 3rd and 6th vertical modes will be discussed in the following sections.

Table 2
VIV events recorded.

Events	U_{South} (m/s)	U_{North} (m/s)	$I_{u,South}$ (%)	$I_{u,North}$ (%)	U_{Mean} (m/s)	$I_{u,Mean}$ (%)	Max acceleration (cm/s^2)	Max Displacement (cm)	Duration (s)
E1	5.42	6.83	6.05	2.09	6.125	4.07	31.36	23.89	210
E2	6.81	7.36	5.24	3.01	7.085	4.125	21.22	16.31	40
E3	5.16	5.78	6.91	4.75	5.47	5.83	9.36	7.9	40
E4	5.75	6.5	6	3.5	6.125	4.75	12.66	9.68	50
E5	4.78	7.17	9.2	1.8	5.975	5.5	20.54	15.85	110
E6	5.32	6.25	8.6	2.4	5.785	5.5	22.37	17.02	120
E7	5.62	7.08	11	2.5	6.35	6.75	13.98	10.3	100
E8	11.2	12.71	6.1	4.2	11.955	5.15	44.06	11.89	40
E9	9.73	10.68	5.88	3.23	10.205	4.555	30.78	9.61	40
E10	10.18	10.56	3.92	2.19	10.37	3.055	18.47	6.48	150
E11	9.75	10.44	7.07	4.9	10.095	5.985	13.46	4.64	50
E12	11.59	11.9	5.65	3.08	11.745	4.365	13.86	7.08	40
E13	10.75	11.16	4.52	2.34	10.955	3.43	17.11	5.54	100
E14	9.96	10.67	4.53	2.54	10.315	3.535	13.85	3.64	30
E15	10.43	11.42	7.44	3.29	10.925	5.365	12.3	4.02	70
E16	10.44	11.33	5.69	4.6	10.885	5.145	50.89	14.78	50
E17	10.2	12	6.01	3.5	11.1	4.755	61.82	17.1	170
E18	10.92	12.64	5.7	3.7	11.78	4.7	40.18	11.39	60
E19	10.53	11.2	4.6	2.5	10.865	3.55	8.46	2.42	130
E20	11.05	11.48	2.9	2.1	11.265	2.5	12.47	2.87	30
E21	9.78	11.51	9.9	3.6	10.645	6.75	25.1	5.05	60
E22	10.3	10.86	4.7	2.3	10.58	3.5	8.4	2.21	90
E23	11.46	11.95	8.6	2.6	11.705	5.6	6.95	1.89	30
E24	10	11.39	11.1	4.3	10.695	7.7	8.74	2.29	30
E25	11.65	12.57	3.6	2.5	12.11	3.05	8.91	2.26	60
E26	11.24	12.17	4.7	2.3	11.705	3.5	13.01	3.03	40
E27	9.87	11.49	3.7	2.2	10.68	2.95	8.15	2.12	90
E28	9.82	10.71	8.1	2.9	10.265	5.5	6.09	1.67	40
E29	10.48	11.08	8.1	4.3	10.78	6.2	15.49	3.7	30
E30	9.74	11.35	8.94	6.72	10.545	7.83	24.96	5.64	150

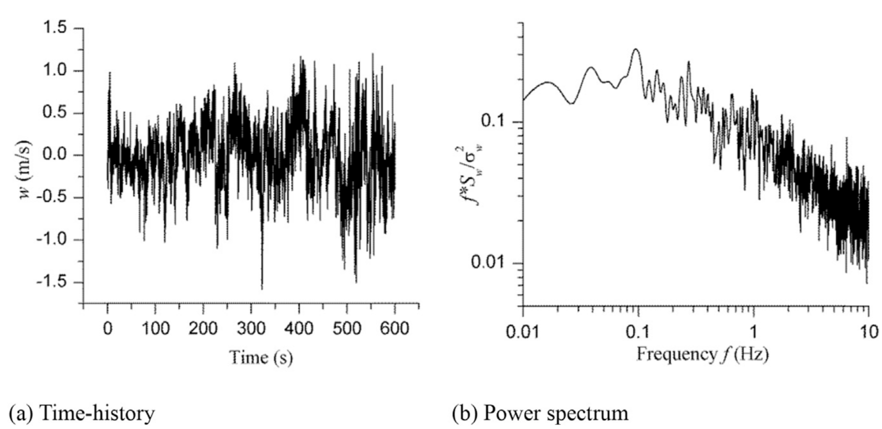


Fig. 4. A sample of the incoming vertical turbulence.

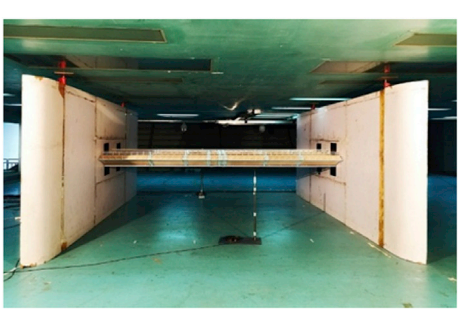


Fig. 5. Mounted sectional model in the wind tunnel.

6.2. Effects of uniformly distributed turbulence on VIV responses

The mode-by-mode VIV analysis is performed for the bridge under uniformly-distributed incoming wind with different turbulence intensities for the 3rd and 6th vertical modes. Fig. 9 shows the effects of the uniformly distributed turbulence on the maximum displacement responses in terms of the analytical results. These responses are computed with the uniformly distributed mean wind speed, which is taken as the mean value of the north and south records. The field-measured maximum displacements are also plotted in Fig. 9 for comparison.

It can be seen from Fig. 9 that the turbulence does not affect the lock-in range of VIV but obviously reduces the maximum displacement responses. The effects of turbulence are more notable on the 3rd vertical mode than on the 6th vertical mode. On the 3rd vertical mode, the

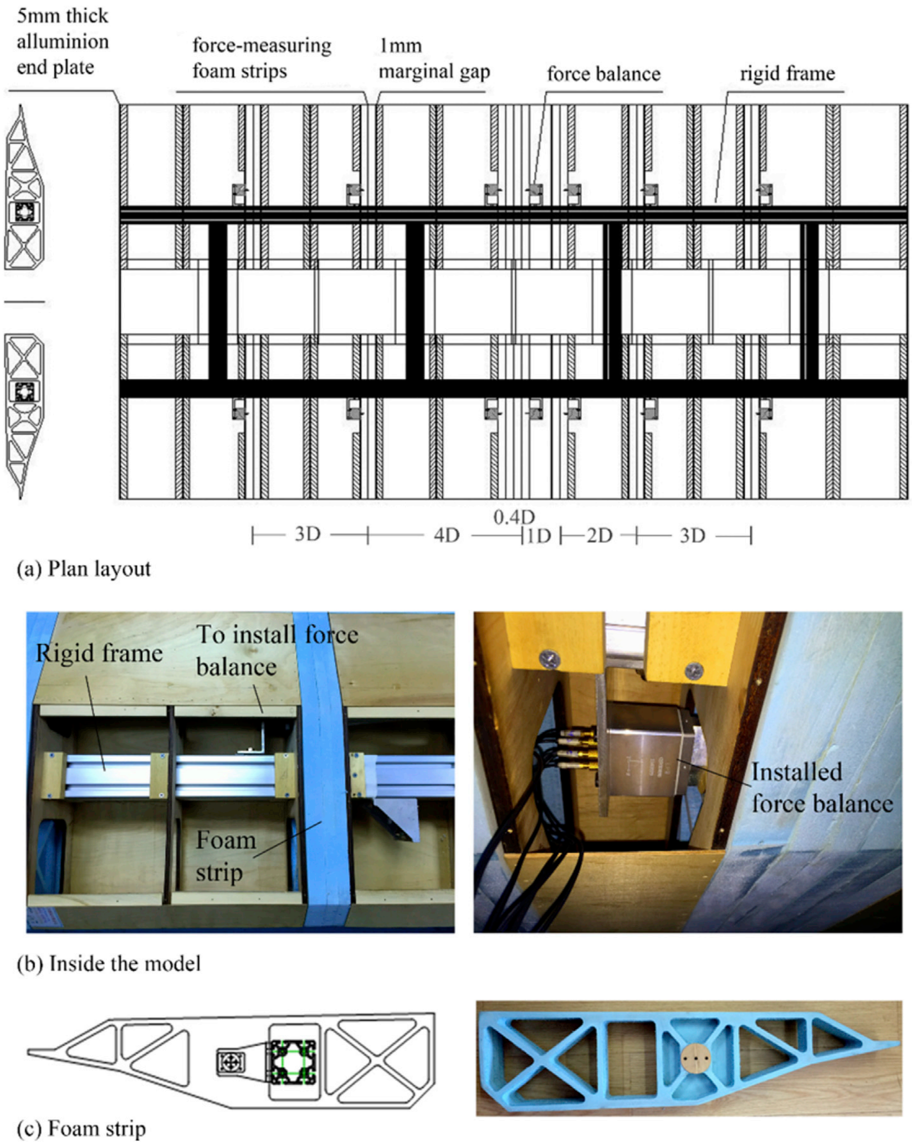


Fig. 6. Plan layout and details of the sectional model.

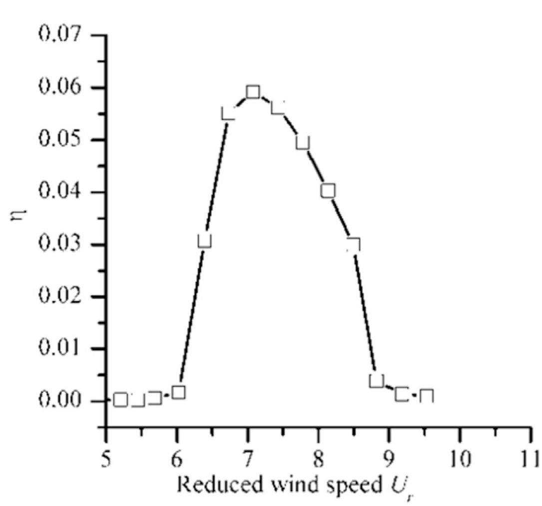


Fig. 7. Lock-in range and maximum displacement responses at different reduced wind speeds.

maximum displacement response reduces about 50% by a vertical turbulence intensity of 6% and the VIV can be totally suppressed by a vertical turbulence intensity of 8%. Meanwhile, a 8% vertical turbulence intensity only reduces the maximum displacement by less than 30% on the 6th vertical mode. This phenomenon indicates that the turbulence has larger effects on lower modes of vibration of the bridge as the energy

Table 3
Identified parameters of the VIF on the Xihoumen bridge deck.

Reduced wind speed	Y_1	Y_2	ε	ε_t
6.026	1.63	3.998	85.7	115.1
6.39	9.95	4.801	161	115.1
6.733	35.10	4.645	285.2	115.1
7.077	50.02	4.726	351.7	115.1
7.428	48.50	4.983	422.6	115.1
7.778	41.10	5.445	593.4	115.1
8.135	30.65	5.883	949.8	115.1
8.493	18.24	6.52	1671	115.1
8.823	1.74	7.104	8464.3	115.1
9.1868	12.50	8.726	4817	115.1

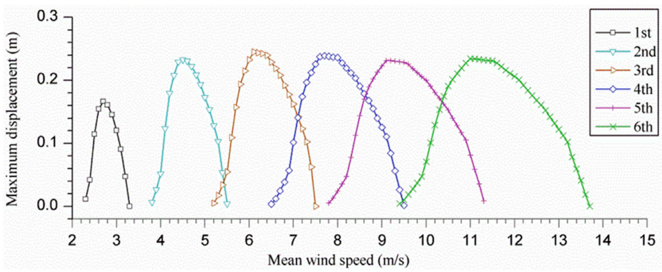


Fig. 8. Vortex-induced maximum displacement responses (uniformly distributed smooth incoming wind).

of turbulence is generally concentrated on the low-frequency range (see Fig. 4). Theoretically speaking, low wind speeds have much higher occurrence probability than high wind speeds, and therefore VIV on lower modes could be more frequently observed. However, for the Xihoumen Bridge, the 6th vertical mode has the most observed VIV events. This could be explained by the postulated fact that turbulence can easily suppress VIV of low-frequency modes.

The analytical maximum displacement responses exhibit a general good consistency with the on-site data for the 3rd vertical mode, as shown in Fig. 9a. However, the analytical maximum displacement responses are much larger than the on-site data for the 6th vertical mode. The VIV events (E9, E11, E16, E17 and E8) generally represent the site-measured maximum displacement curve in the lock-in range for the 6th vertical mode, as shown in Fig. 9b. However, the maximum displacement of most other events falls far below this curve, indicating the VIVs in these events are not fully developed. This implicates a possibility that it is difficult for the VIV to fully develop on high-order modes as high wind speed cannot sustain long enough.

It should be noted that the analytical results presented above do not take account of the non-uniform distribution of the incoming wind. The effects of the non-uniformity and the comparison between the analytical and site-measured maximum displacements considering the non-uniformity are presented and discussed in the next subsection.

6.3. Effects of the non-uniformity of incoming wind on VIV responses

VIV events (E1 and E17) are the events of the largest vortex-induced responses on the 3rd and 6th vertical modes, respectively. The reduced mean wind speeds of these two events are very close to those of the largest vortex-induced responses in the analytical results. As a result, these two events are selected for analyses of the effects of non-uniformity of incoming wind.

Five cases are analyzed for each event: (1) only uniformly distributed mean wind speed is considered for the bridge; (2) only non-uniformly distributed mean wind speed is applied; (3) the uniformly distributed mean wind speed is applied together with uniformly distributed turbulences; (4) the non-uniformly distributed mean wind speed is applied together with the uniformly distributed turbulences; (5) the non-uniformly distributed mean wind speed is applied together with the non-uniformly distributed turbulences. The uniformly distributed mean wind speed is taken as the mean value of the north and south records. The non-uniformly distributed mean wind speed is assumed to be varying linearly based on the north and south measured data. Similarly, the uniformly distributed turbulence harbors the mean intensity of the north and south records while the non-uniformly distributed turbulence has a linearly varying intensity based on the two measured data.

Fig. 10 shows the analytical VIV time-histories of the event E1 for the five cases. A comparison between Fig. 10a and b shows that the non-uniform distribution of mean wind speed reduces the maximum displacement. The comparisons between Fig. 10a and c as well as between Fig. 10b and d shows that turbulence not only reduces the maximum displacement but also introduces fluctuation in vibration amplitude. Comparison between Fig. 10d and e shows that the effects of the non-uniform distribution of turbulence on the maximum displacement are negligible. It shall be pointed out that although the mean intensity of the north and south turbulent wind records is used to define the uniformly distributed turbulence, the incoming turbulence is actually non-steady, as shown in Fig. 4a. The time-varying characteristics of the incoming turbulence cause the displacement response of the bridge to vary, as shown in Fig. 10c–e.

Fig. 11 shows the analytical VIV time-histories of the event E17 for the five cases. The observations here are very similar with those of the event E1. First, the non-uniform distribution of mean wind speed reduces the maximum displacement. Second, turbulence not only reduces the maximum displacement but also introduces fluctuation in vibration amplitude. Third, the effects of the non-uniform distribution of turbulence on the maximum displacement are negligible.

Fig. 10 shows that the analytical maximum displacement predicted for the case of the uniform distribution of mean wind speed without turbulence is very close to the site-measured value. With the reduction effect of turbulence and non-uniformly distributed wind, the analytical maximum displacement deviates from the site-measured value. On the contrary, Fig. 11 shows that the analytical maximum displacement predicted for the case of the non-uniform distribution of mean wind

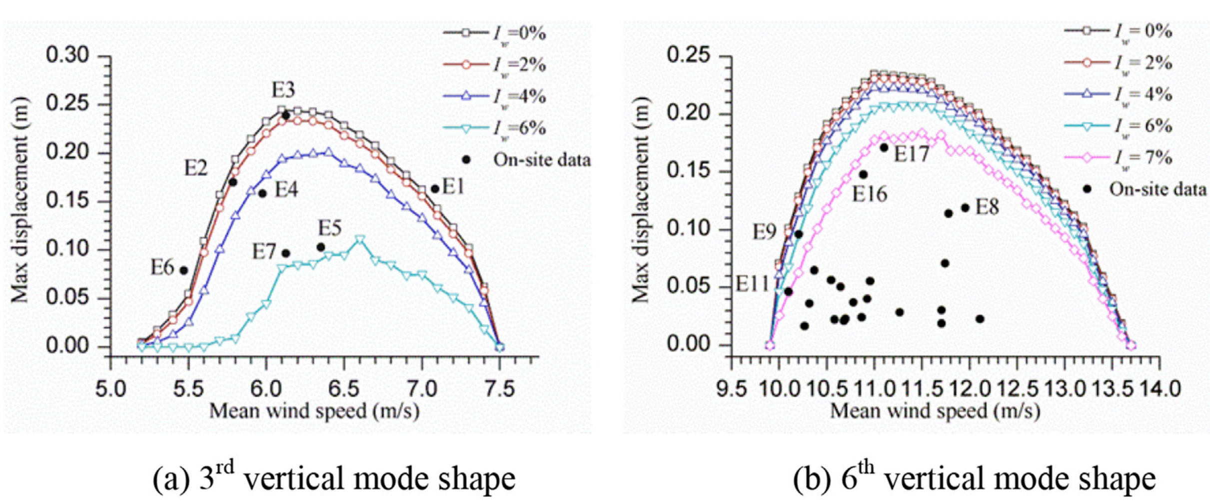
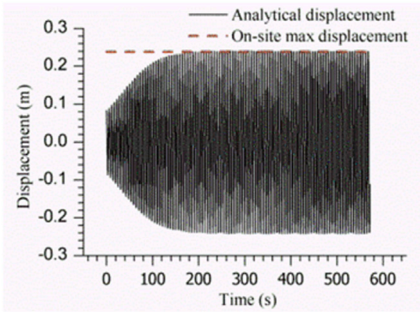
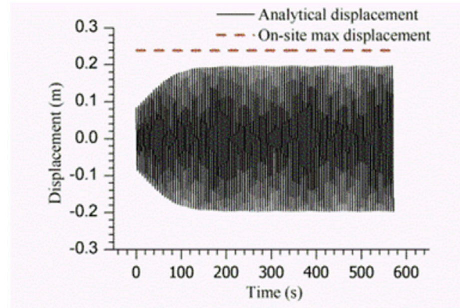


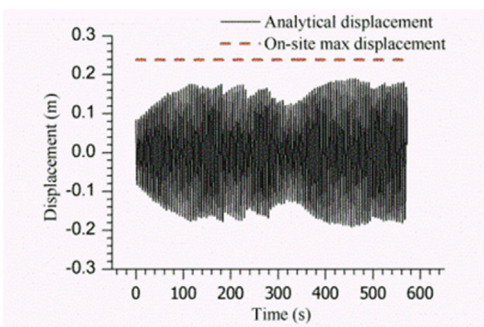
Fig. 9. Effects of turbulence and comparison between analytical and field-measurement responses.



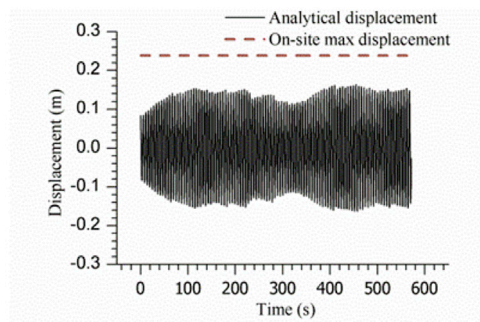
(a) uniform mean wind speed without turbulence



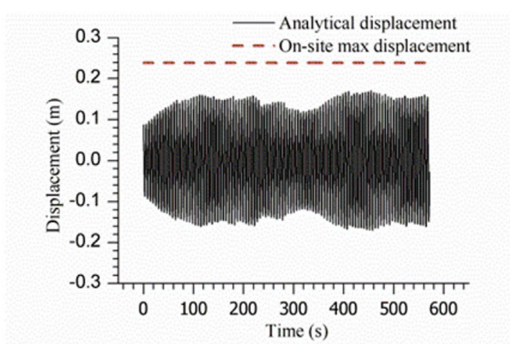
(b) non-uniform mean wind speed without turbulence



(c) uniform mean wind speed with uniform turbulence



(d) non-uniform mean wind speed with uniform turbulence



(e) non-uniform mean wind speed with non-uniform turbulence

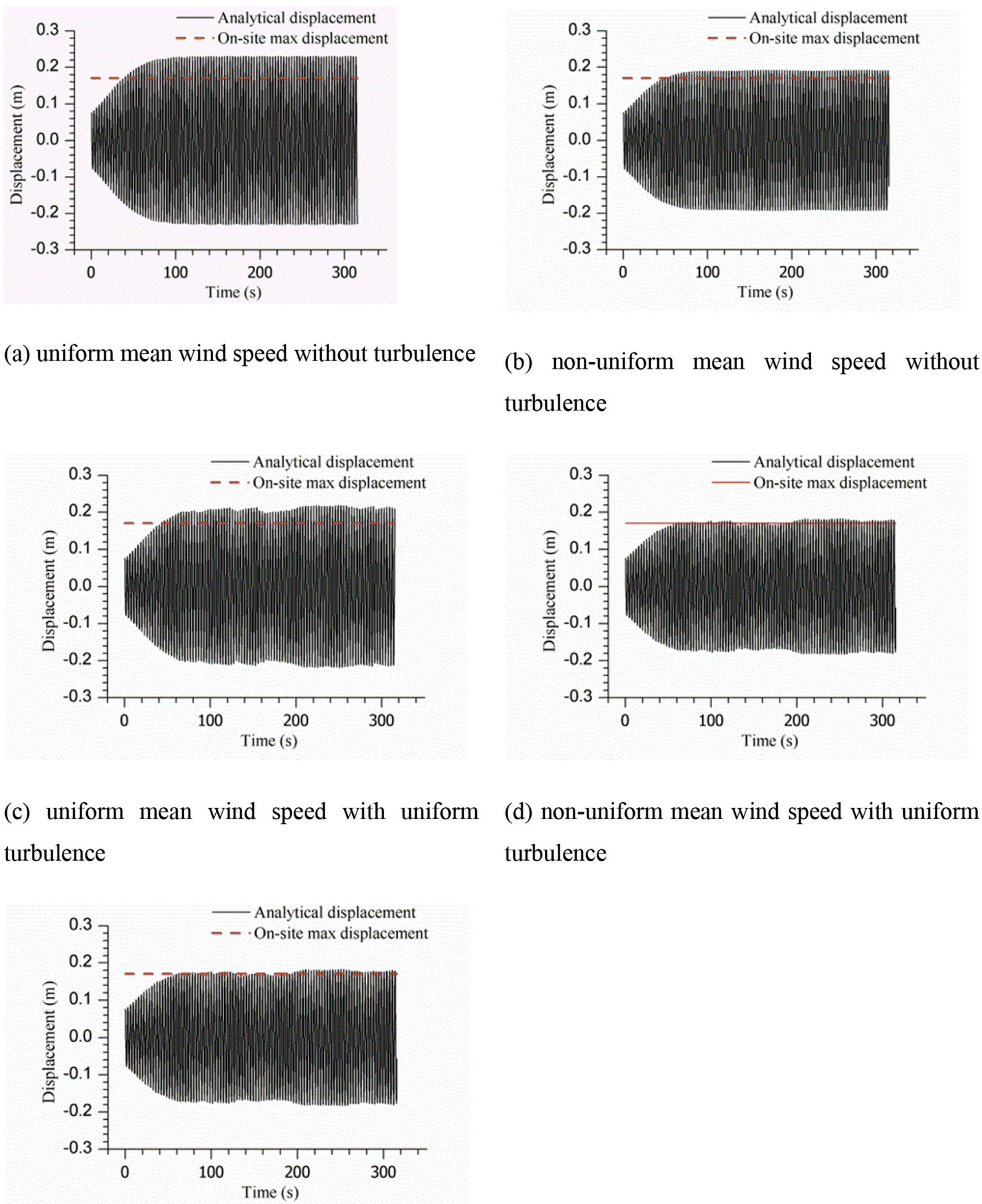
Fig. 10. Analytical VIV time-histories of event E1 for different cases.

speed with uniform or non-uniform turbulence is very close to the site-measured value. In another word, the analytical results show a good consistency with the on-site data on the 6th vertical mode while underestimate the responses for the 3rd vertical mode. While the actual reason of this finding is still under investigation, it is suspected that this could result from the differences between the linear damping assumed by the analyses and the actual nonlinear damping of the prototype bridge.

Fig. 12 shows the analytical VIV time-histories of the event E4 and event E16, neither of which is the event of the maximum response. The main conclusions here are very similar with those of the event E1 and E17. The non-uniformity of the incoming wind, both the non-uniform mean wind speed and the turbulence, reduces the VIV response. In both cases, the consideration of the non-uniformity diminishes the differences between the analytical and measured responses.

7. Conclusions

A mode by mode VIV analysis method has been developed in this study for long-span bridges with twin-box decks under non-uniformly



(a) uniform mean wind speed without turbulence

(b) non-uniform mean wind speed without turbulence

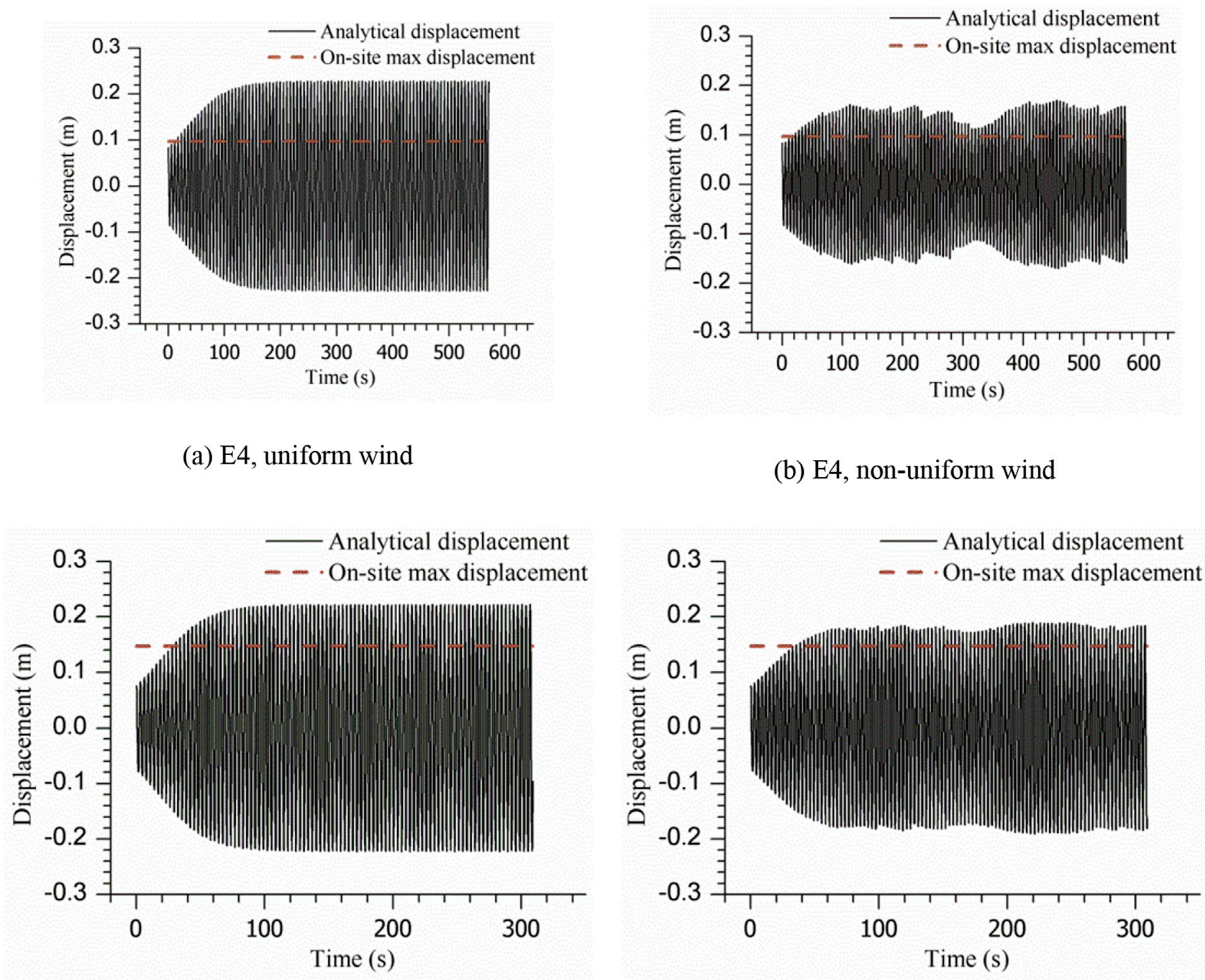
(c) uniform mean wind speed with uniform turbulence

(d) non-uniform mean wind speed with uniform turbulence

(e) non-uniform mean wind speed with non-uniform turbulence

Fig. 11. Analytical VIV time-histories of event E17 for different cases.

distributed mean and turbulent winds along the bridge axis. The developed method has been applied to a long-span suspension bridge that suffered VIV. The analytical VIV results have been compared with the on-site VIV events recorded by the health monitoring system installed on the bridge. The main conclusions that can be drawn from the results are as follows.



(a) E4, uniform wind

(b) E4, non-uniform wind

(a) E16, uniform wind

(a) E16, non-uniform wind

Fig. 12. Analytical VIV time-histories of events E4 and E16.

(1) The comparative results show that the proposed VIV analysis method can be used to predict the VIR of long-span bridges with a twin-box deck.
(2) Turbulence reduces the VIV, and the reduction effect is larger on the VIV responses with low-order modes than those with high-order modes as the energy of turbulence is concentrated on the low-frequency range. This suggests that turbulence may easily suppress VIV on low-frequency modes.
(3) The non-uniform distribution of mean wind speed reduces the VIV responses. For long-span bridges, neglecting the non-uniform distribution of wind speed may overestimate the VIV responses. However, the effects of the non-uniform distribution of turbulence are negligible.

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